Lezione 4
Introduction to FPGAs and Xilinx devices

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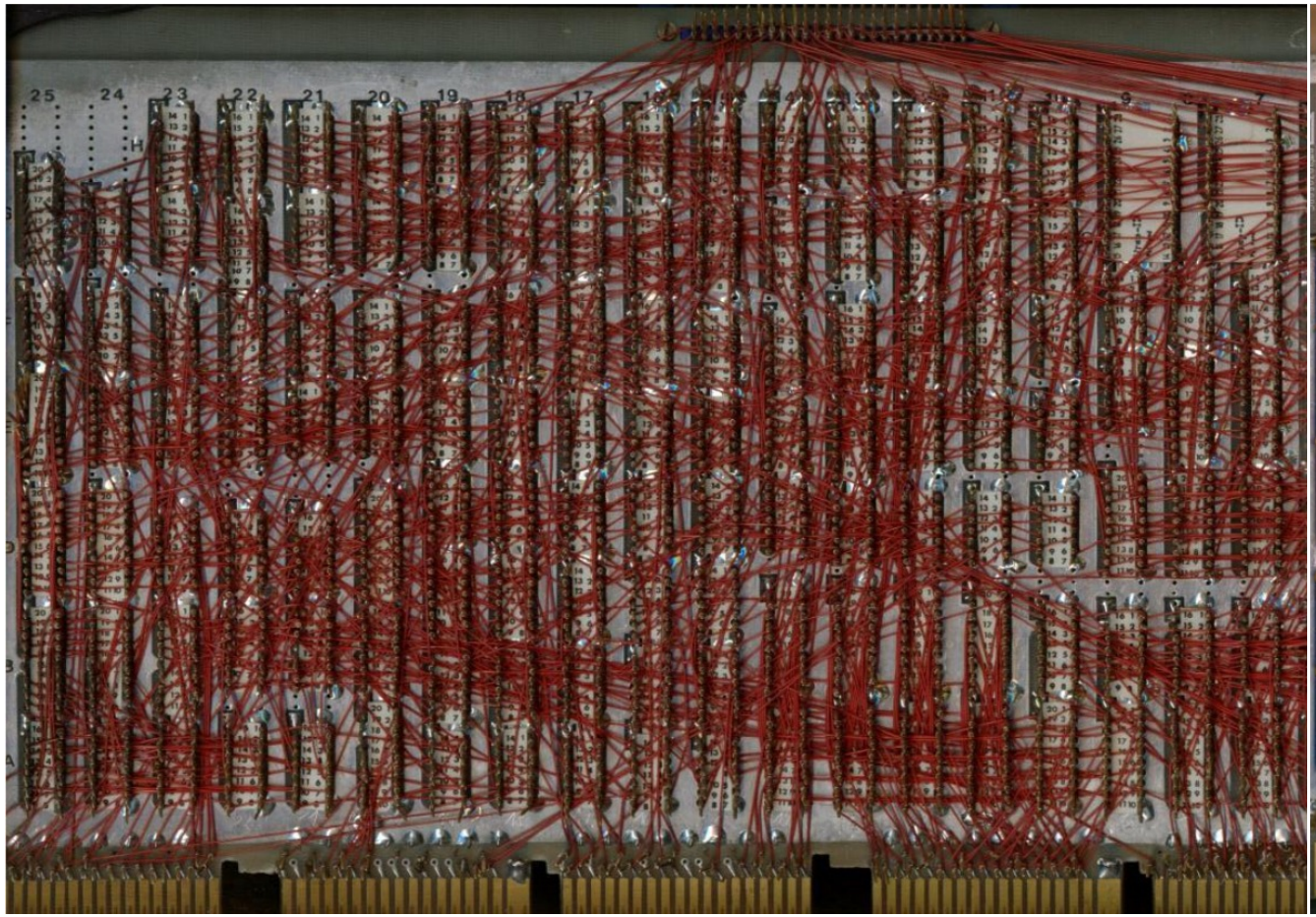
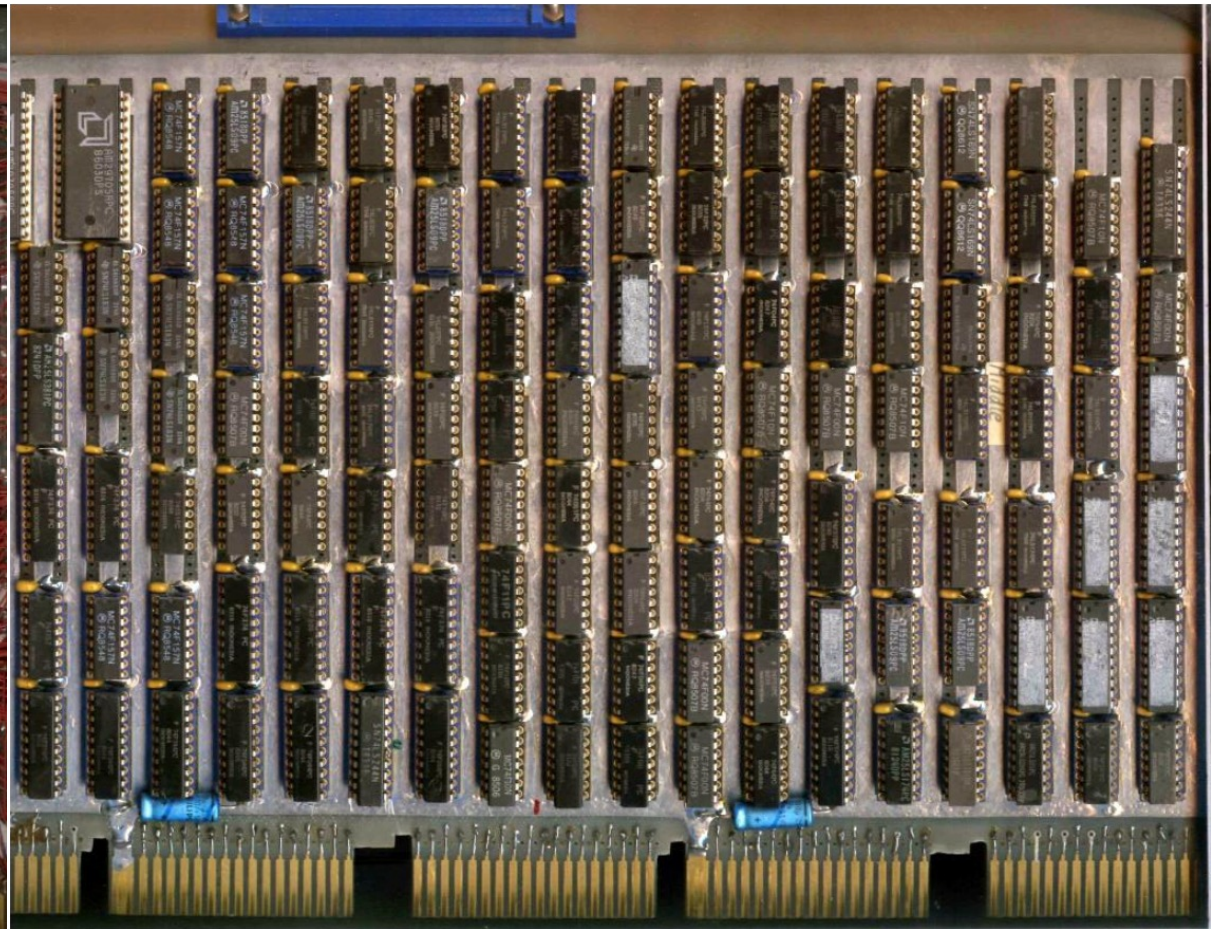
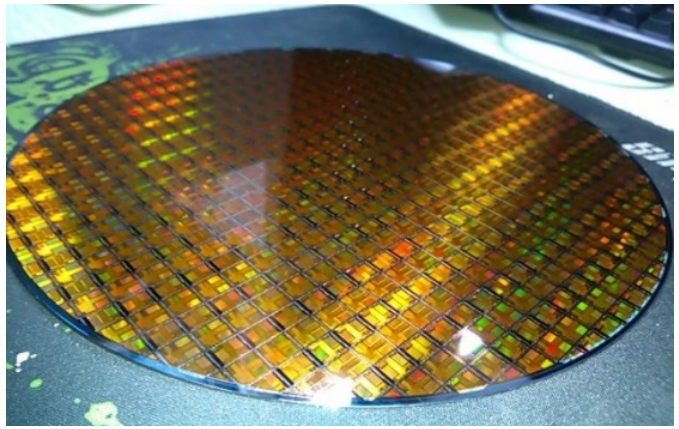
Early times of digital electronics

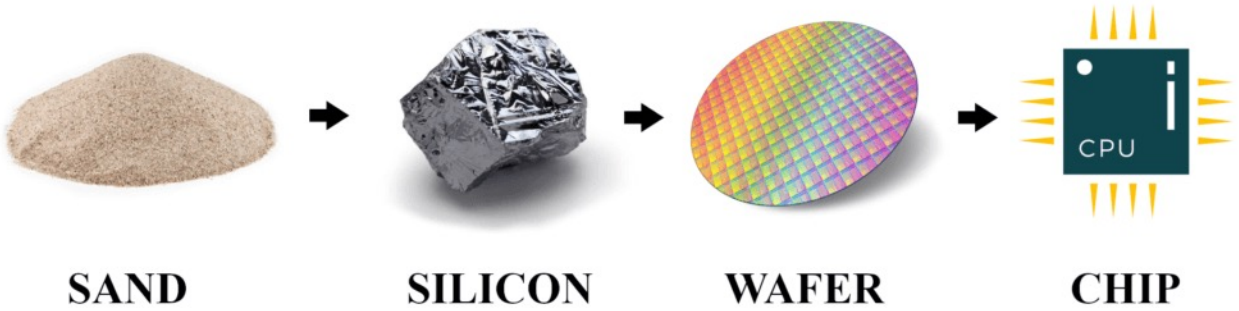




Why programmable ICs?

- Producing a Chip costs much more than producing a PCB
- Using one Chip on many PCBs increases Number of Chips and reduces Costs per Chip
- If a Chip is programmable, it can be used on many PCBs in different situations
- Producing one wafer costs ~ 10k€
  - Not included manpower to design wafer & packaging
- Saving space on a PCB





What are FPGAs?

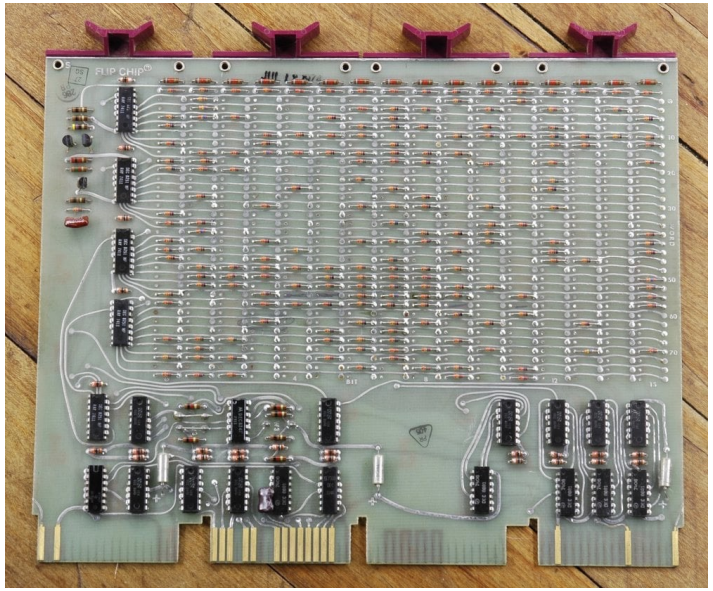
Field **P**rogrammable **G**ate **A**rray

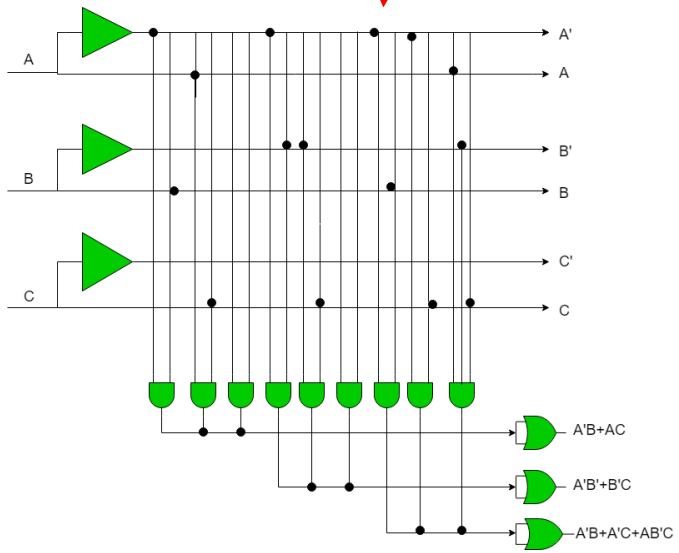
- Simple:
  - Programmable Logic Devices

- More detailed:
  - Programmable Logic Devices
    - With lot of additional function-block (e.g. RAM, DSP)
    - Flexible configuration
    - Can host microcontroller

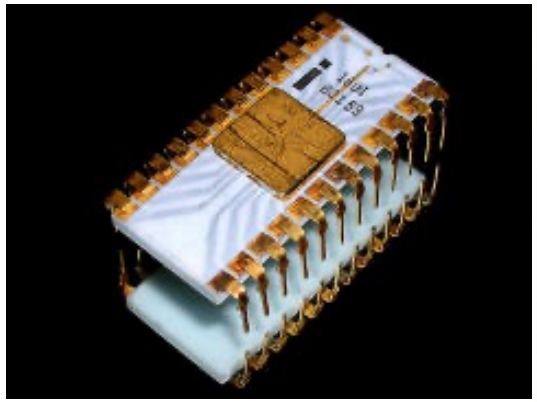
History of Programmable Logic Devices /1

- **1960s:** Fuse Configurable Diode Matrix
  - First form of programmable logic
  - One-Time Programmable (OTP)
- **1971:** Programmable ROM (**PROM**)
  - Logic functions stored as truth tables
  - Still limited flexibility and efficiency
- **PAL** (Programmable Array Logic):
  - Fixed AND/OR logic structure
  - Programmable interconnections
  - OTP technology



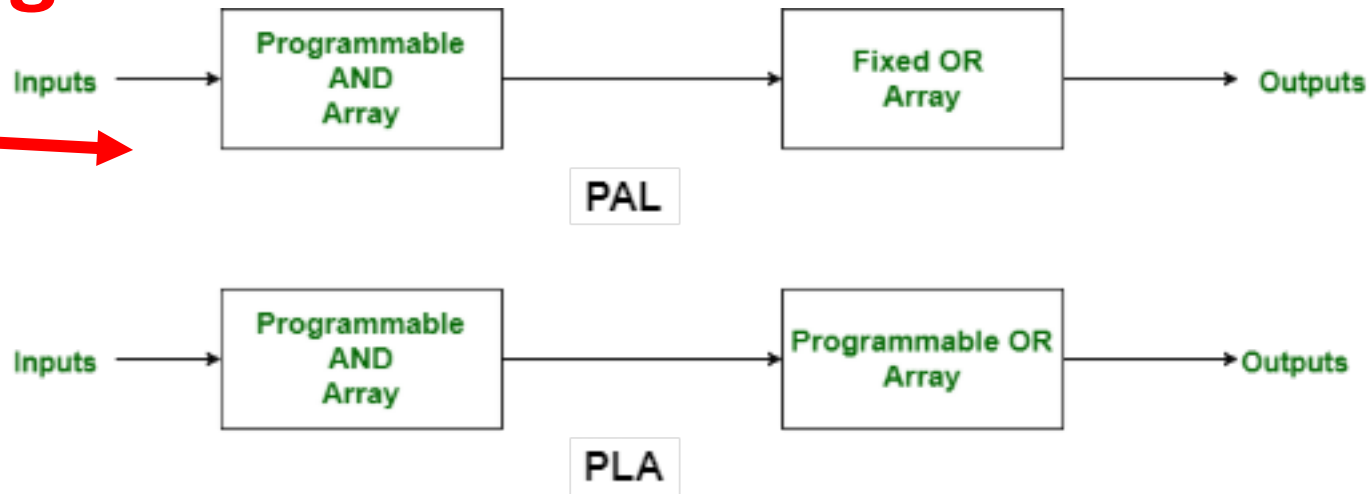


256 x 8-bit Static/Dynamic PROM



History of Programmable Logic Devices /2

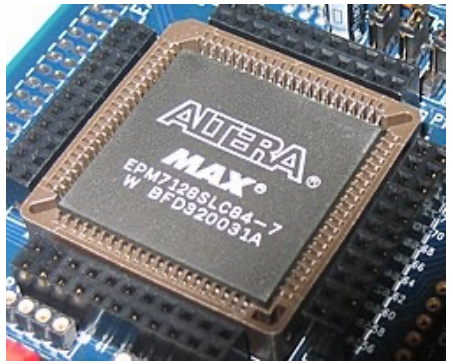
- **1978: PLA (Programmable Logic Array)**
  - Programmable AND plane, fixed OR plane
  - Faster and cheaper than PLA
- **CPLD (Complex Programmable Logic Device)**
  - Multiple PAL-like blocks
  - In-system programmable
  - Configuration **retained after power-off**



The most noticeable difference between a large CPLD and a small FPGA is the presence of on-chip non-volatile memory in the CPLD, which allows CPLDs to be used for bootloader functions

Comparison with other Programmable Logic Devices

- PLA has a programmable AND gate array and programmable OR gate array.
- PAL has a programmable AND gate array but a fixed OR gate array.
- ROM has a fixed AND gate array but programmable OR gate array.

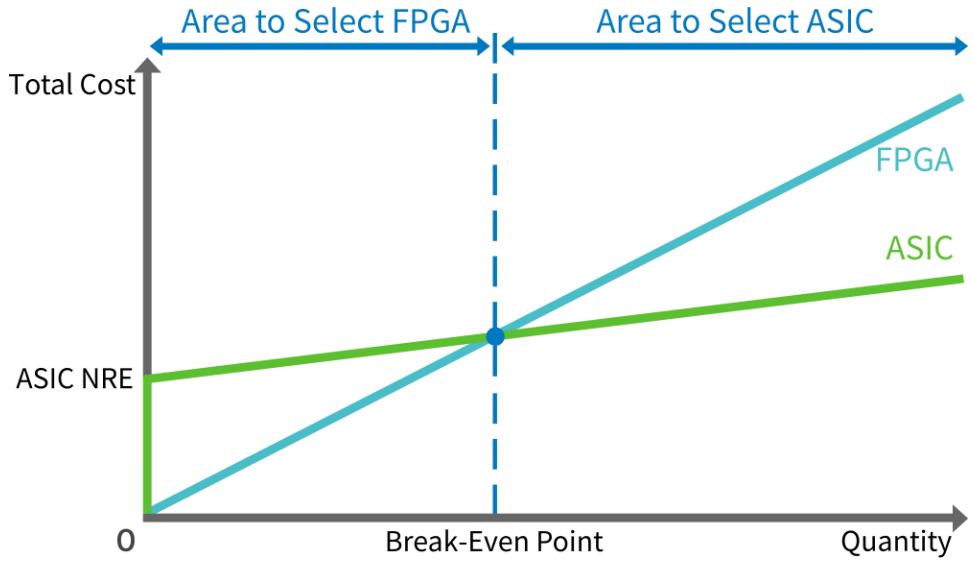


History of Programmable Logic Devices /3

- **1989:** First Field Programmable Gate Array (FPGA)
  - Configurable Logic Blocks (CLBs)
  - Programmable routing fabric
  - Massive parallelism
- Compared to CPLDs:
  - Much higher logic density
  - SRAM-based configuration (loaded at startup)
  - Ideal for complex systems

Price point

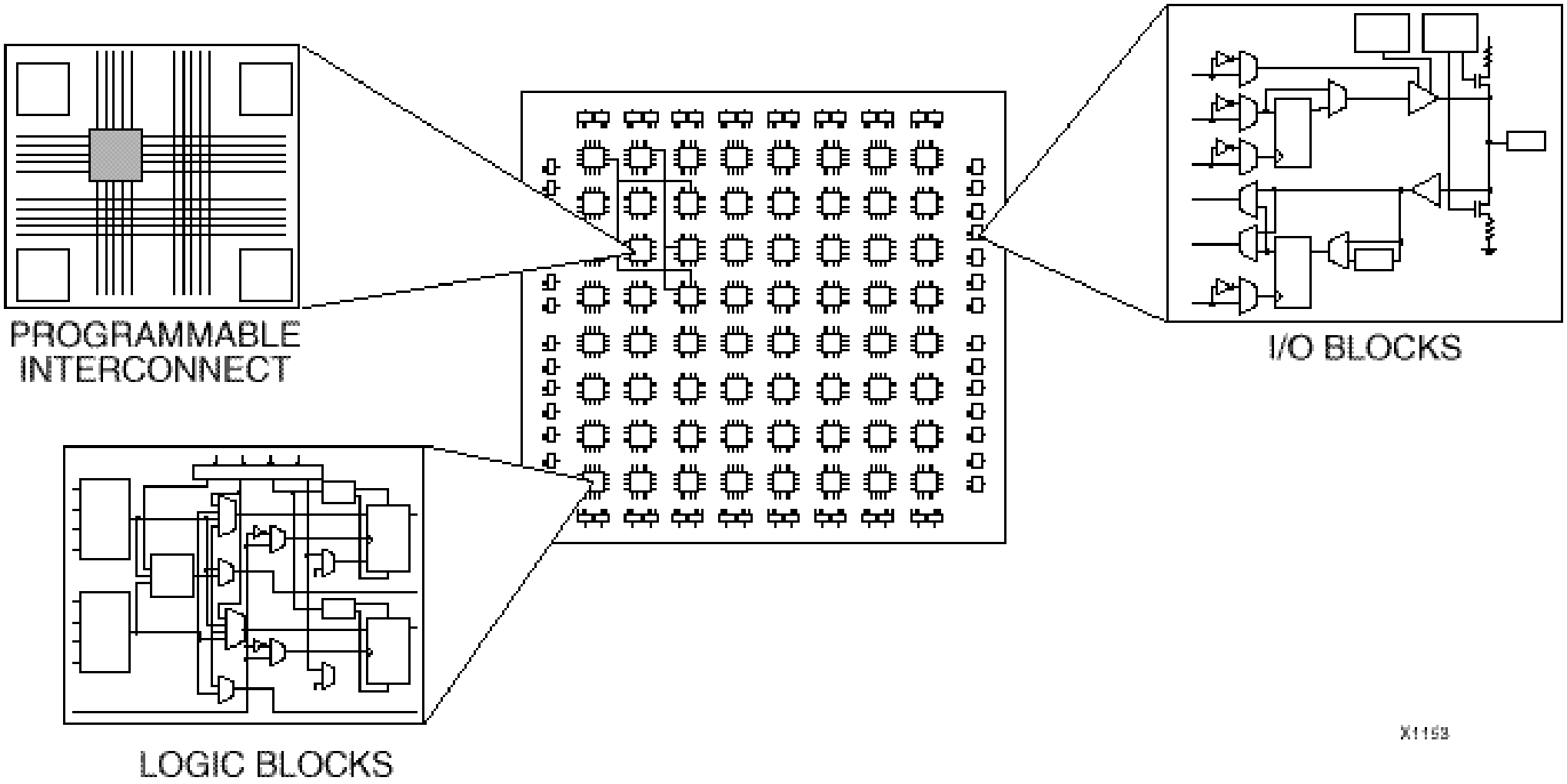
	Produttore Codice prodotto ICE40UL640-CM36AI **Codice Mouser** 842-ICE40UL640CM36AI	Lattice	FPGA - Array di gate programmabile su campo FPGA iCE40-UltraLite 1.2V CBGA PKG Maggiori informazioni	Scheda dati	243 A magazzino Confezione alternativa	1 2,58 € 25 2,24 € 100 2,12 €
	Produttore Codice prodotto ICE40UL1K-CM36AI **Codice Mouser** 842-ICE40UL1K-CM36AI	Lattice	FPGA - Array di gate programmabile su campo FPGA iCE40-UltraLite 1.2V CBGA PKG Maggiori informazioni	Scheda dati	722 A magazzino Confezione alternativa	1 2,67 € 25 2,32 € 100 2,24 €
	Produttore Codice prodotto ICE40LP384-CM49 **Codice Mouser** 842-ICE40LP384CM49	Lattice	FPGA - Array di gate programmabile su campo iCE40LP Ultra LowPwr 384 LUTs 1.2V Maggiori informazioni	Scheda dati	836 A magazzino Confezione alternativa	1 2,67 € 25 2,32 € 100 2,24 €
	Produttore Codice prodotto ICE40LP384-CM36TR **Codice Mouser** 842-ICE40LP384CM36TR	Lattice	FPGA - Array di gate programmabile su campo iCE40LP Ultra LowPwr 384 LUTs 1.2V Maggiori informazioni	Scheda dati	Tempo di consegna, se non a magazzino 24 settimane Confezione alternativa	Nastro continuo 20.000 2,68 €
	Produttore Codice prodotto AGIA035R39A2I3V **Codice Mouser** 989-AGIA035R39A2I3V	Altera	FPGA - Array di gate programmabile su campo	Scheda dati	Tempo di consegna diretta 16 settimane	3 38.140,14 €
	Produttore Codice prodotto 1ST210EU2F50I1VG **Codice Mouser** 989-1ST210EU2F50I1VG	Altera	FPGA - Array di gate programmabile su campo	Scheda dati	Tempo di consegna diretta 16 settimane	3 37.496,86 €
	Produttore Codice prodotto AGIA035R39A2E2VB **Codice Mouser** 989-AGIA035R39A2E2VB	Altera	FPGA - Array di gate programmabile su campo	Scheda dati	Tempo di consegna diretta 16 settimane	3 37.080,62 €
	Produttore Codice prodotto AGIA035R39A2E3E **Codice Mouser** 989-AGIA035R39A2E3E	Altera	FPGA - Array di gate programmabile su campo	Scheda dati	Tempo di consegna diretta 16 settimane	3 37.080,62 €
	Produttore Codice prodotto XCVU7P-L2FLVB2104E **Codice Mouser** 217-VU7P-L2FLVB2104E	AMD / Xilinx	FPGA - Array di gate programmabile su campo XCVU7P-L2FLVB2104E EAR	Scheda dati	Tempo di consegna, se non a magazzino 37 settimane	1 38.475,97 €
	Produttore Codice prodotto XCVU7P-2FLVB2104I **Codice Mouser** 217-CVU7P-2FLVB2104I	AMD / Xilinx	FPGA - Array di gate programmabile su campo XCVU7P-2FLVB2104I EAR	Scheda dati	Tempo di consegna, se non a magazzino 37 settimane	1 38.475,97 €



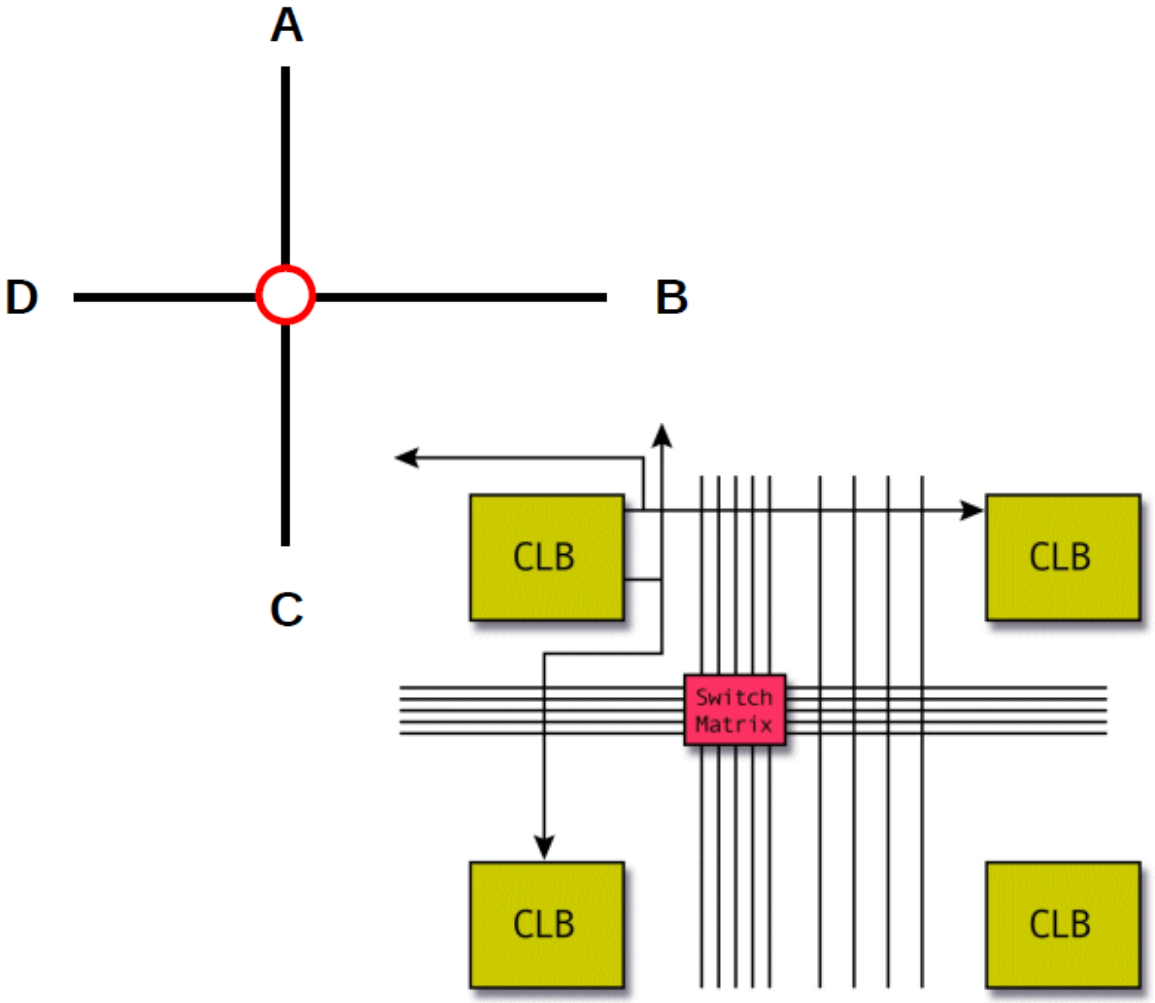
Comparing FPGA and Dedicated ICs

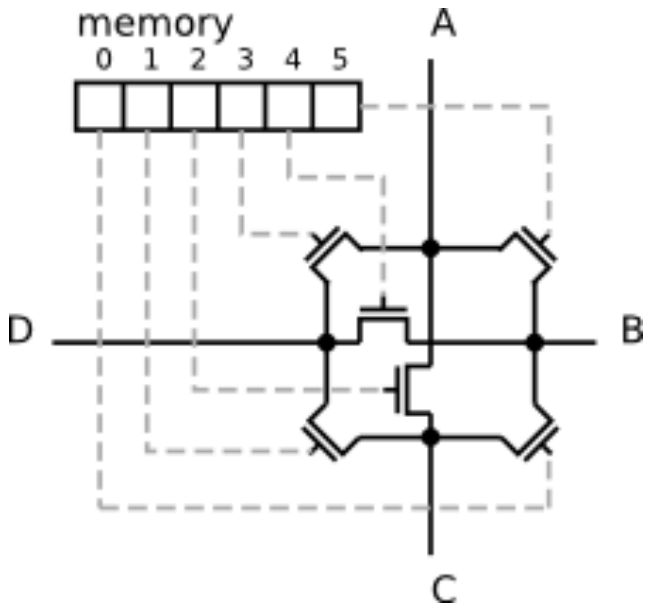
	FPGA	Dedicated IC
Up-front cost	Smaller	Larger
Per-unit cost	Larger	Smaller
Time to market	Shorter	Longer
Speed	Slower	Faster
Power consumption	Greater	Smaller
Update in the field	Straightforward	Very difficult
Density	Lesser	Greater
Design Flow	Simpler	More complex
Granularity	Logic blocks	Individual cells
Need for gate-level verification	Little	Much
Technology upgrade path	Easier	Harder
Additional features	Many	Fewer

Fundamental FPGA components

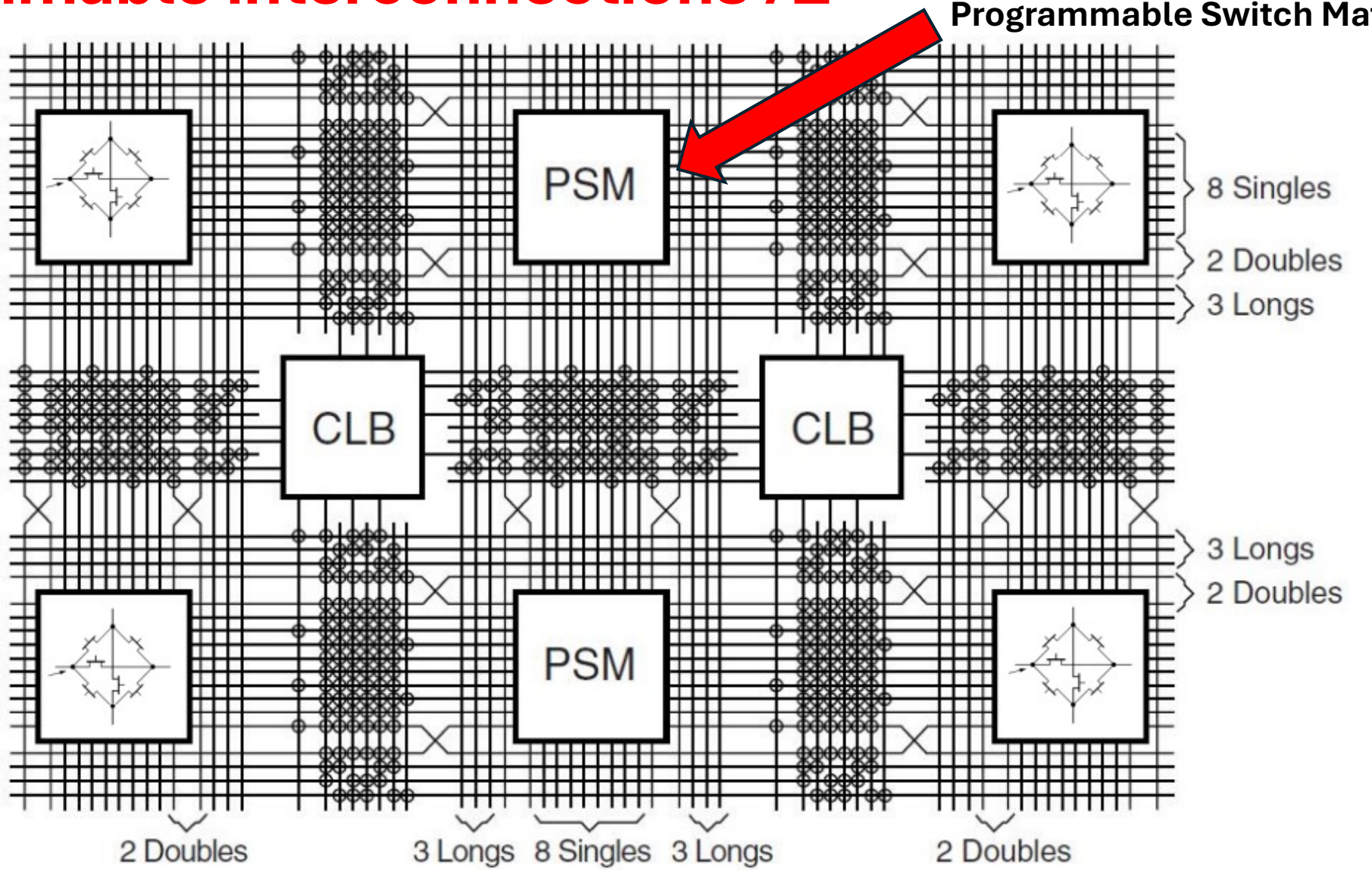


Programmable interconnections /1



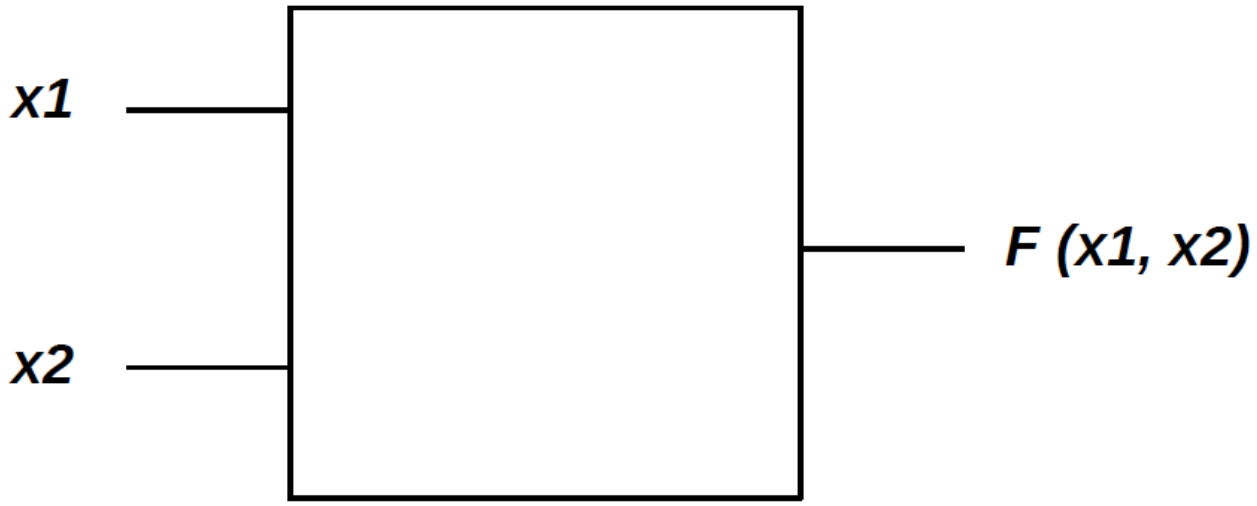


Programmable interconnections /2



Look-up tables (LUT) /1

How to implement a 2-bit truth table (boolean logic) ?

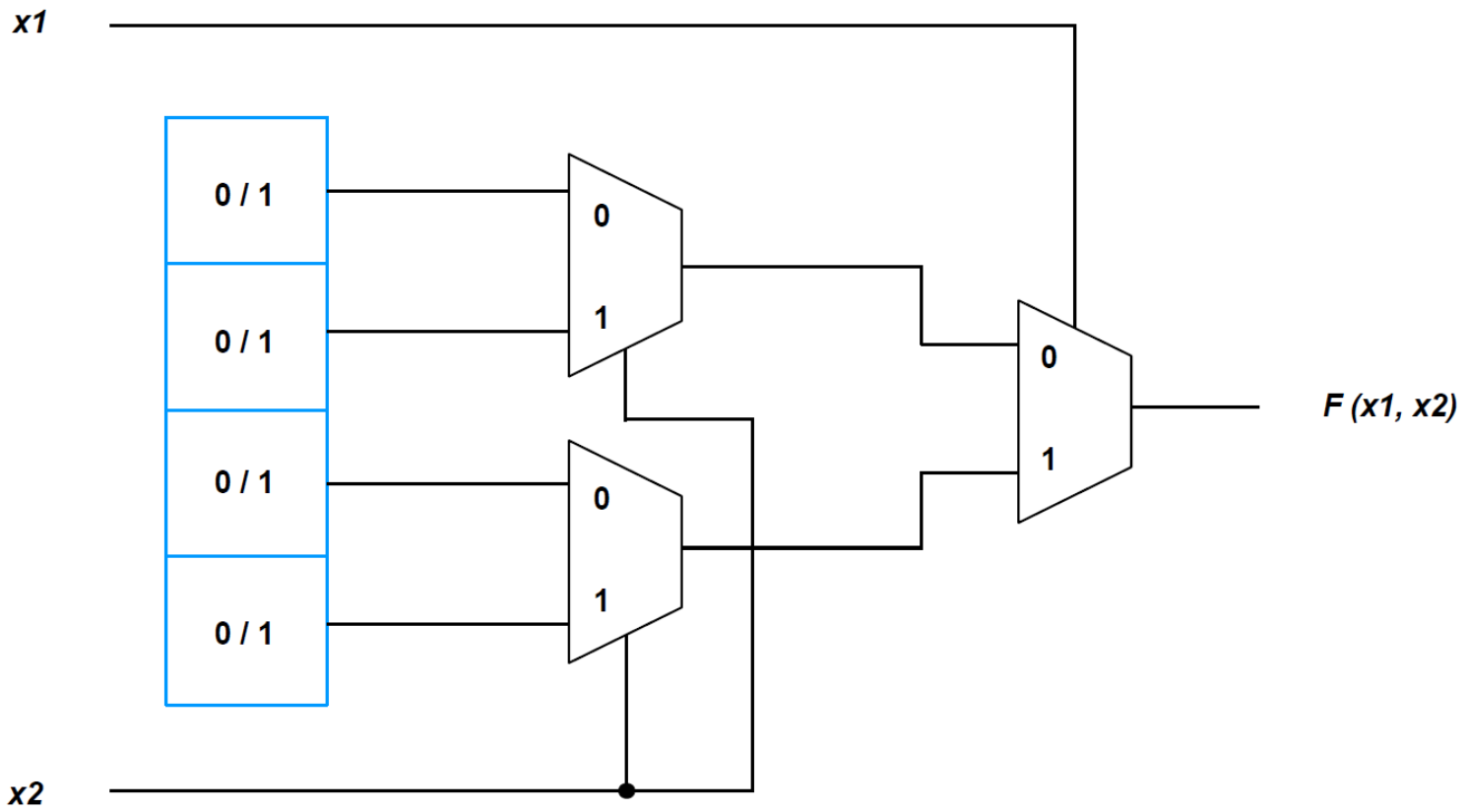


XNOR

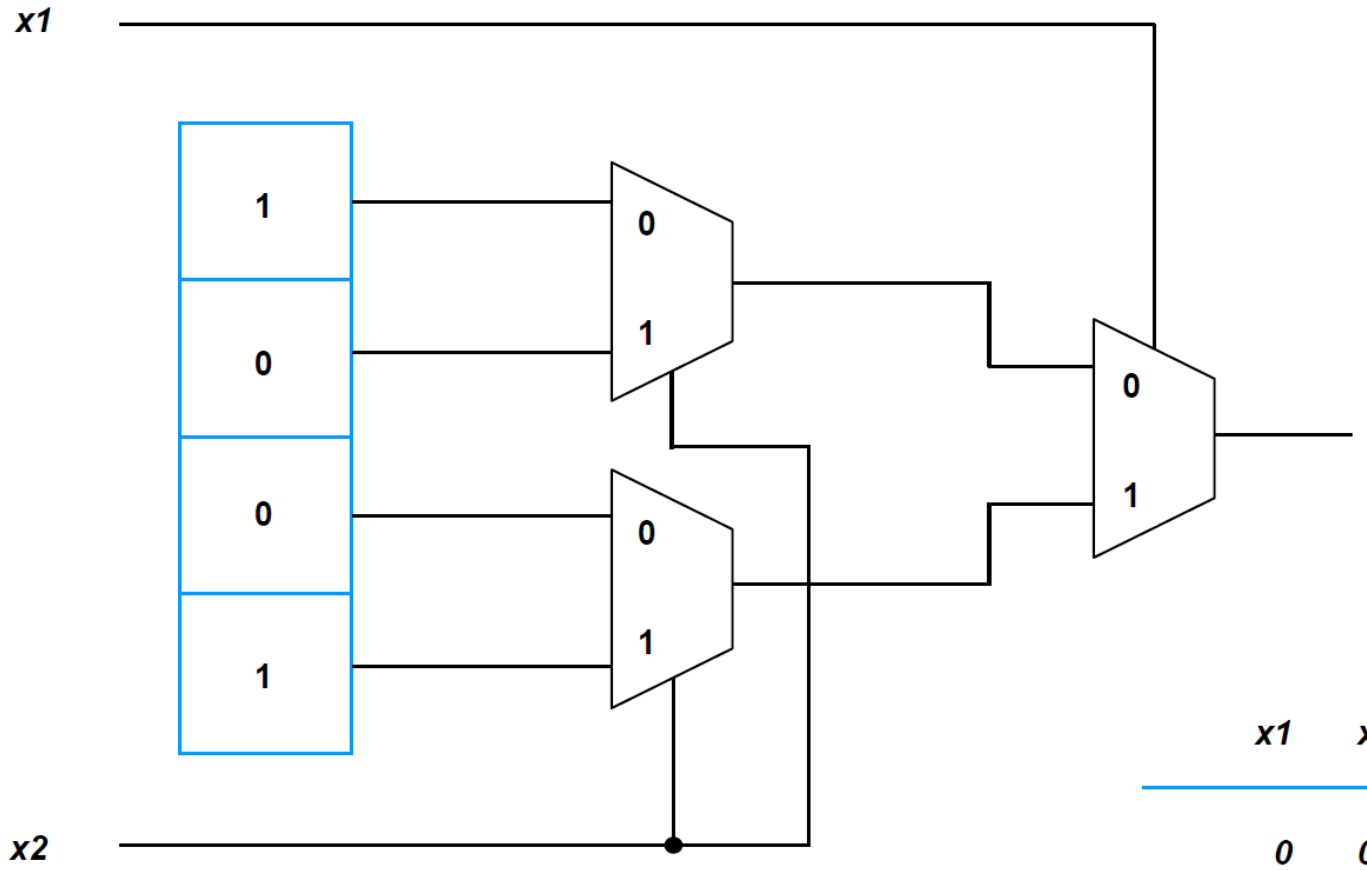
x1	x2	F
0	0	1
0	1	0
1	0	0
1	1	1

Look-up tables (LUT) /2

FPGA solution: write the truth table into a RAM !



Look-up tables (LUT) /3



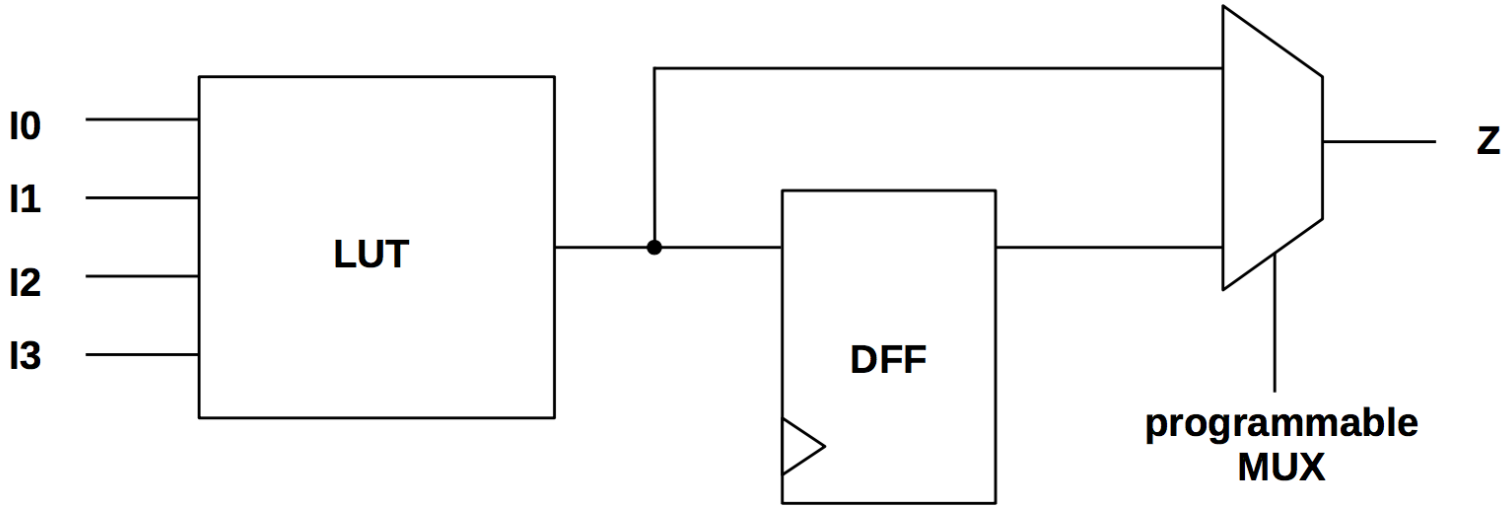
x1	x2	F
0	0	1
0	1	0
1	0	0
1	1	1

Logic Cell (LC)

Xilinx terminology :

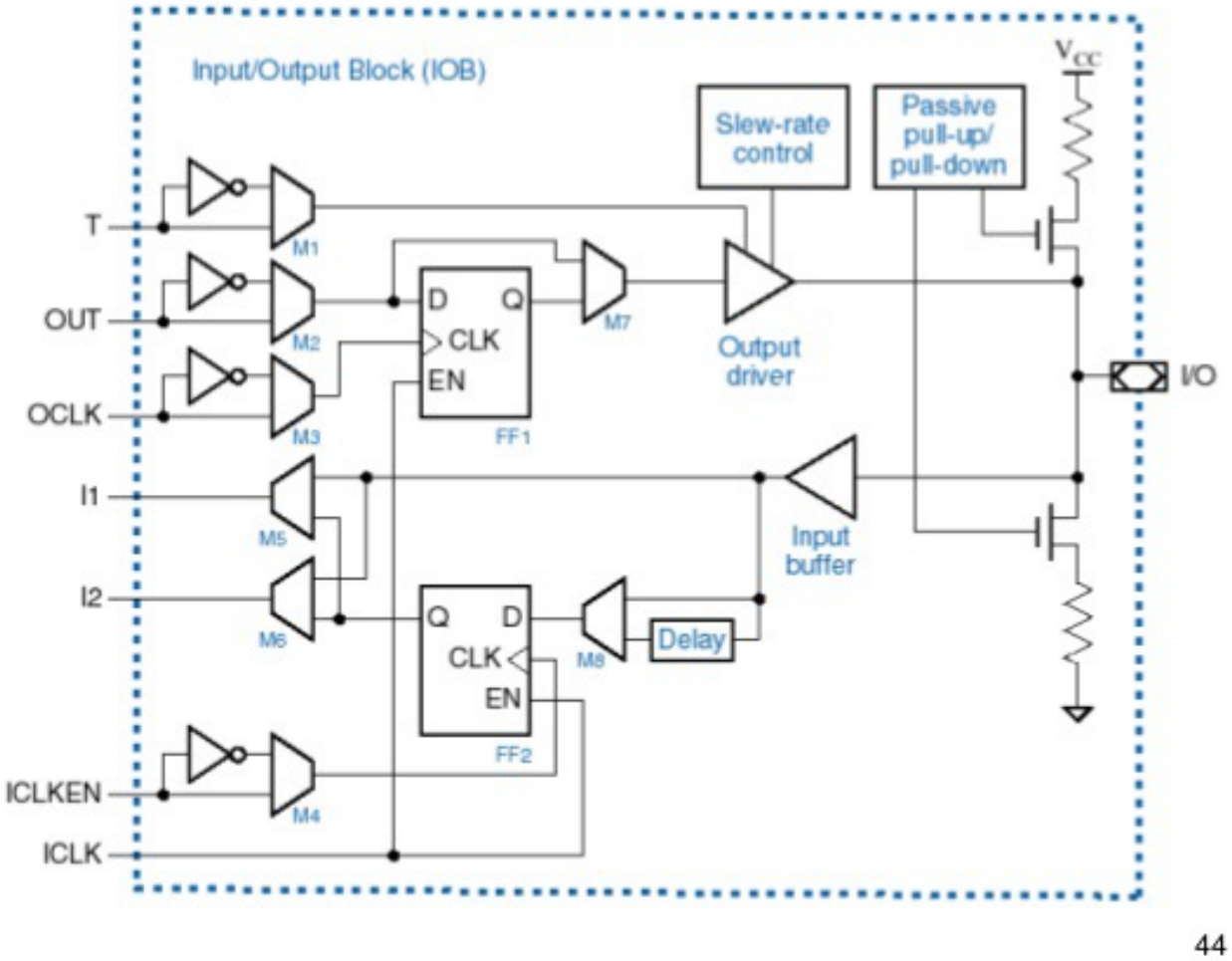
- **Logic Cell (LC)** = 1x LUT + 1 register (FF) + programmable MUX
- **LUT** implemented as 6-inputs, 64-bit RAM **WHY?**
- slice = 4x LUTs + 8 registers (FFs) + MUXs + fast-carry chain (CLA)
- **Congurable Logic Block (CLB)** = 2x slices
- Basic Element (BEL) = any device primitive

carry-lookahead adder



Programmable I/O pads

I/O BLOCK



Programmable pad

Xilinx FPGA line-up

Xilinx 7-series family comparison

Different size, number of programmable logic cells, I/O capabilities, performance and cost depending on the chosen device family :

- Spartan 7 : lowest-cost, lowest-power, good I/O performance, lowest number of resources
- Artix 7 : low-power, medium number of resources and I/O, good choice for DSP
- Kintex 7 : best compromise between number of resources, I/O, speed, power and cost
- Virtex 7 : highest system performance (resources, speed, I/O) but also largest power consumption and highest cost

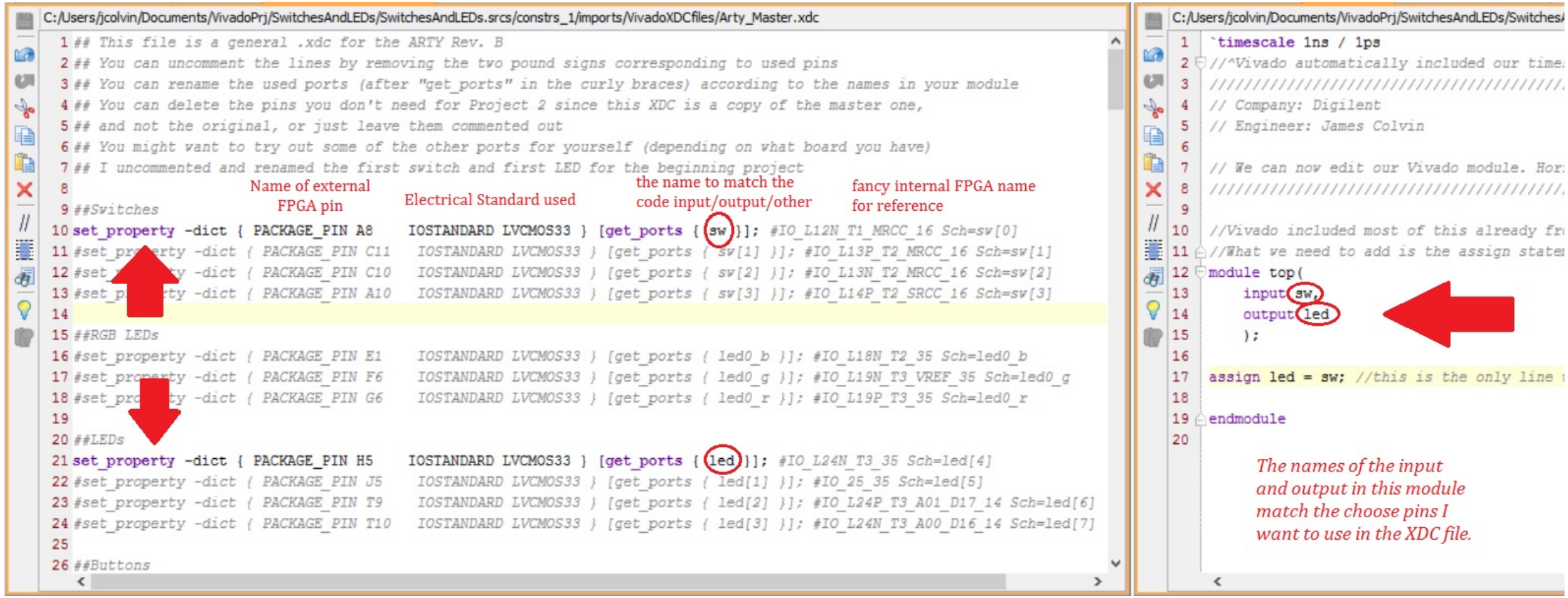
Max. Capability	Spartan-7	Artix-7	Kintex-7	Virtex-7
Logic Cells	102K	215K	478K	1,955K
Block RAM	4.2 Mb	13 Mb	34 Mb	68 Mb
DSP Slices	160	740	1,920	3,600
DSP Performance	176 GMAC/s	929 GMAC/s	2,845 GMAC/s	5,335 GMAC/s
MicroBlaze CPU	260 DMIPs	303 DMIPs	438 DMIPs	441 DMIPs
Transceivers	–	16	32	96
Transceiver Speed	–	6.6 Gb/s	12.5 Gb/s	28.05 Gb/s
Serial Bandwidth	–	211 Gb/s	800 Gb/s	2,784 Gb/s
PCIe Interface	–	x4 Gen2	x8 Gen2	x8 Gen3
Memory Interface	800 Mb/s	1,066 Mb/s	1,866 Mb/s	1,866 Mb/s
I/O Pins	400	500	500	1,200
I/O Voltage	1.2V–3.3V	1.2V–3.3V	1.2V–3.3V	1.2V–3.3V
Package Options	Low-Cost, Wire-Bond	Low-Cost, Wire-Bond, Bare-Die Flip-Chip	Bare-Die Flip-Chip and High-Performance Flip-Chip	Highest Performance Flip-Chip

How “programming” a FPGA?

Hardware Description HDL	Describe what the hardware should do.
Syntax Check	Check Syntax
Synthesis	“compiles” the design to transform HDL source into an architecture-specific design netlist. (connections)
Translate	Merges incoming netlists and constraints into a Xilinx design file → connections are merged with timing
Map	Fits the design into the available resources → tell the FPGA which gate structures should be used
Place & Route	Places and routes the design to the timing constrains → decide, which gate is used for which function
Programming	Write configuration into the FPGA or configuration memory

Xilinx XDC – Xilinx Design Constraints file

XDC constraints are based on the standard **Synopsys Design Constraints (SDC)** format. SDC has been in use and evolving for more than 20 years, making it the most popular and proven format for describing design constraints.



Recommended Constraints Sequence

```
## Timing Assertions Section # Primary clocks
# Virtual clocks
# Generated clocks # Clock Groups
# Bus Skew constraints
# Input and output delay constraints
```

```
## Timing Exceptions Section # False Paths
# Max Delay / Min Delay # Multicycle Paths
# Case Analysis # Disable Timing

## Physical Constraints Section
# located anywhere in the file, preferably before or after the timing
constraints # or stored in a separate constraint file
```

Example: Digilent Arty-A7 evaluation board

Device: Xilinx Artix-7